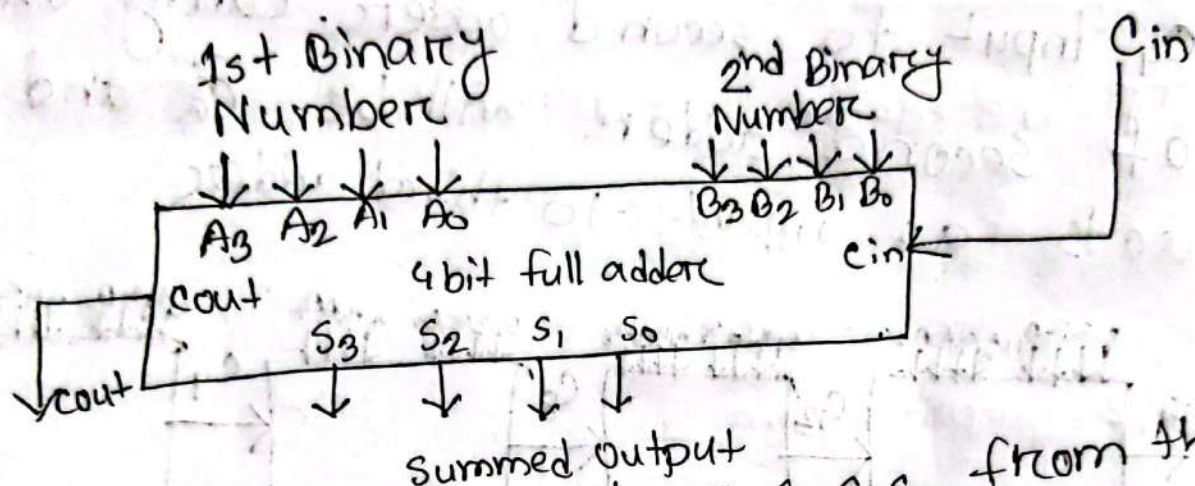


chapter - 5

1. Excess - 3 code we get by adding decimal 3 to BCD. so we here we will reverse the procedure to get BCD code out of excess 3 code input. Let excess 3 input $A_3A_2A_1A_0$



So to get BCD code $S_3S_2S_1S_0$ from this excess 3 code will subtract 3 from it. we will carry out this subtraction using two's complement method.

$$A_3A_2A_1A_0 - (3)_{10} = A_3A_2A_1A_0 + \text{1s complement of } 3_{10} + 1$$

$$= A_3A_2A_1A_0 + (1100)_2 + 1_2$$

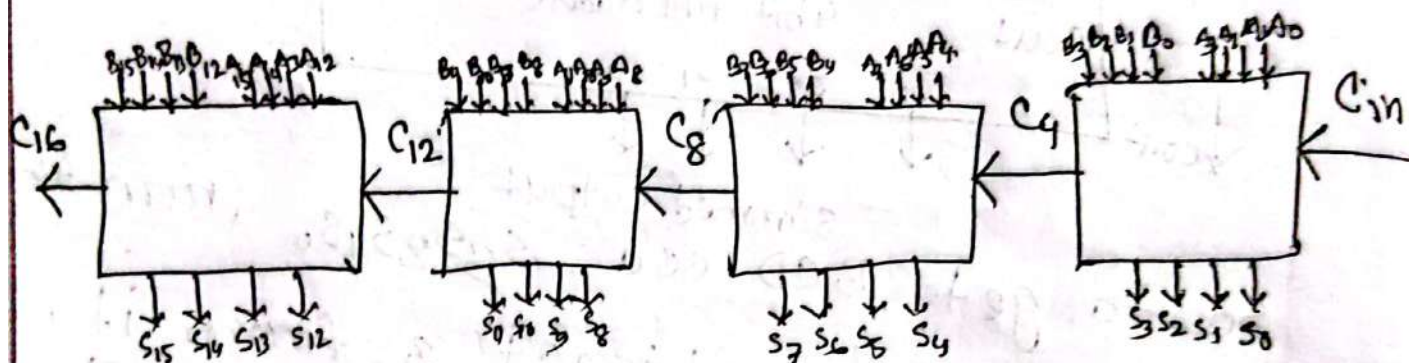
So in adder we assign

$$B_3B_2B_1B_0 = (1100)_2$$

$$\text{And } C_{in} = (1)_2$$

And the output $S_3S_2S_1S_0$ will give BCD code

2. First 4 bit adder (0 to 3) bits, second adder add (4 to 7) bits, third adder adds bits and final 4th adder adds bits. Carry generated out of first adder will be C_4 input to second adder. Carry out of second adder will be C_8 and acts as input to third adder.



3. Using 4 exclusive-OR gates and 4-bit Full adders MSI circuits, construct a 4 bit parallel adder/subtractor. Use an input select variable V so that when $V=0$ the circuit adds and when $V=1$ the circuit subtracts.

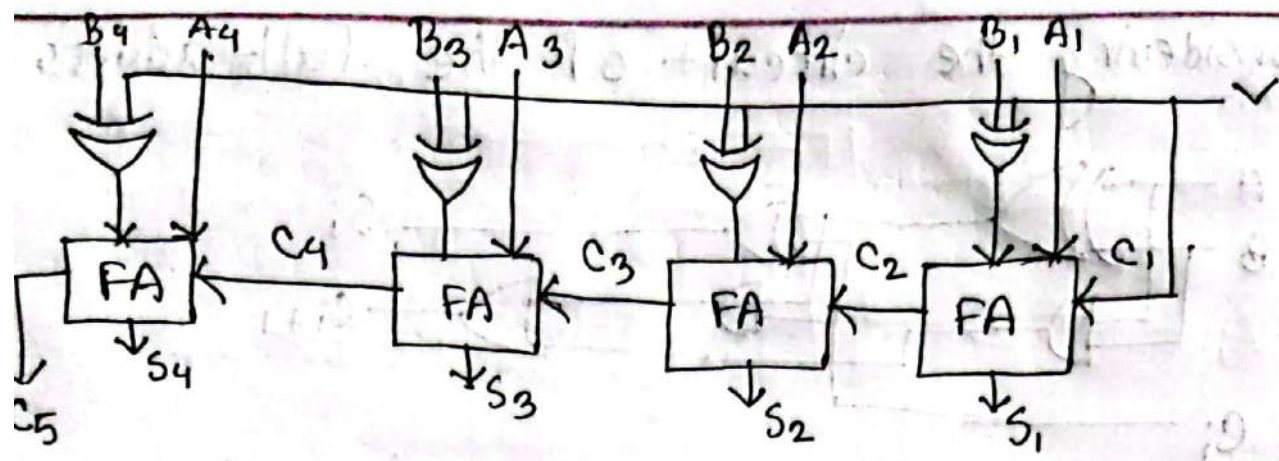


Fig-4-bit adders subtractors

The adders subtractors circuit of the figure has the following values for mode input V and data inputs A and B .

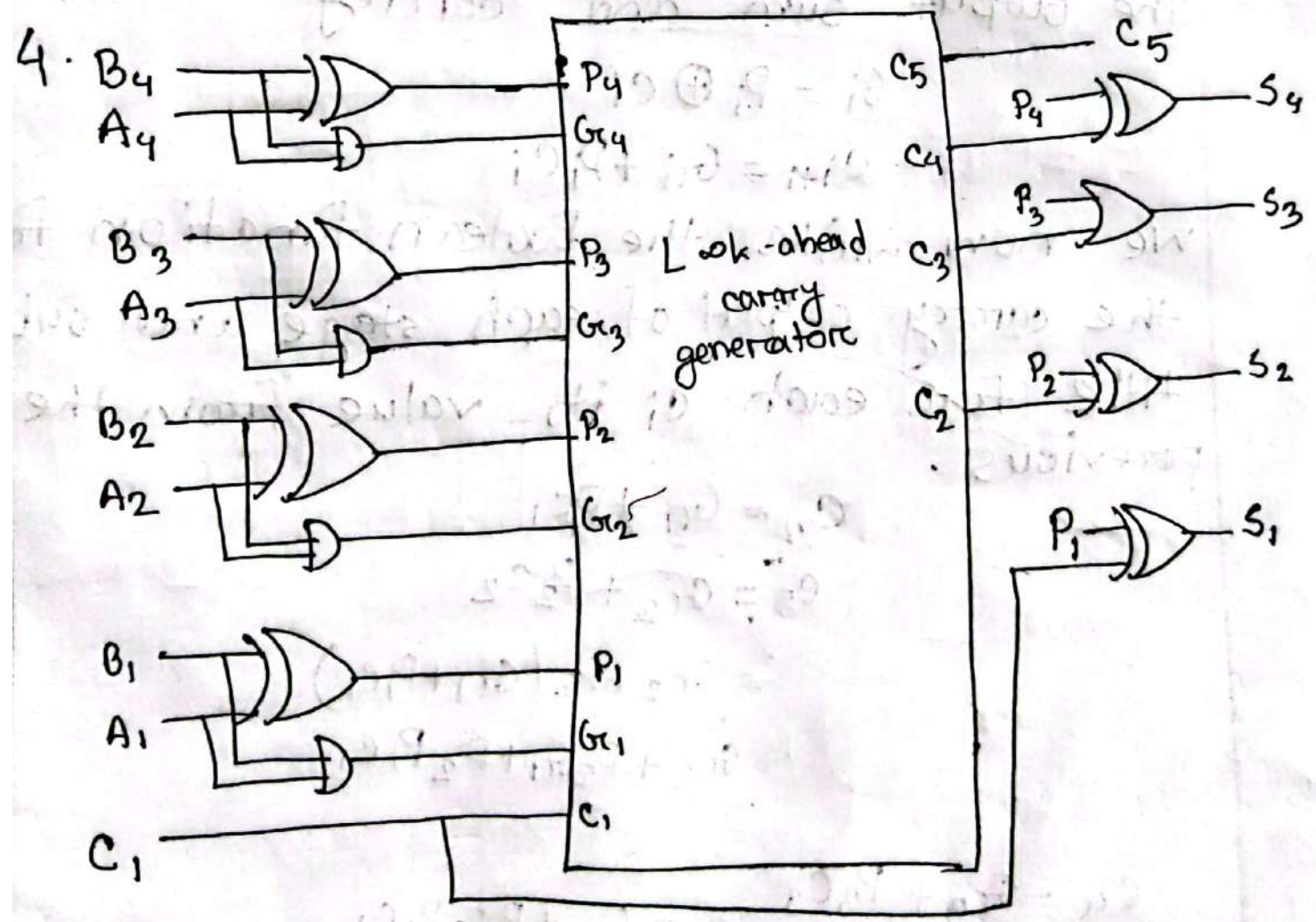
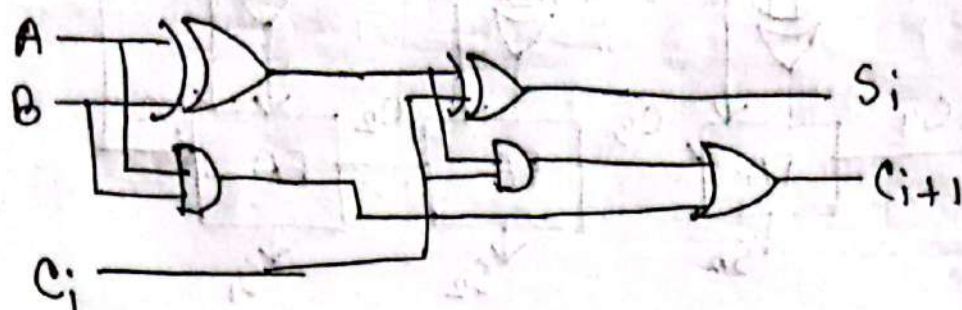


Fig-4-bit full adders

Considering the circuit of the full-adders



If we define two new binary

$$P_i = A_i \oplus B_i$$

$$G_i = A_i B_i$$

the output sum and carry

$$S_i = P_i \oplus C_i$$

$$C_{i+1} = G_i + P_i C_i$$

we now write the Boolean function for the carry output of each stage and substitute for each C_i its value from the previous.

$$C_2 = G_1 + P_1 C_1$$

$$C_3 = G_2 + P_2 C_2$$

$$= G_2 + P_2 (G_1 + P_1 C_1)$$

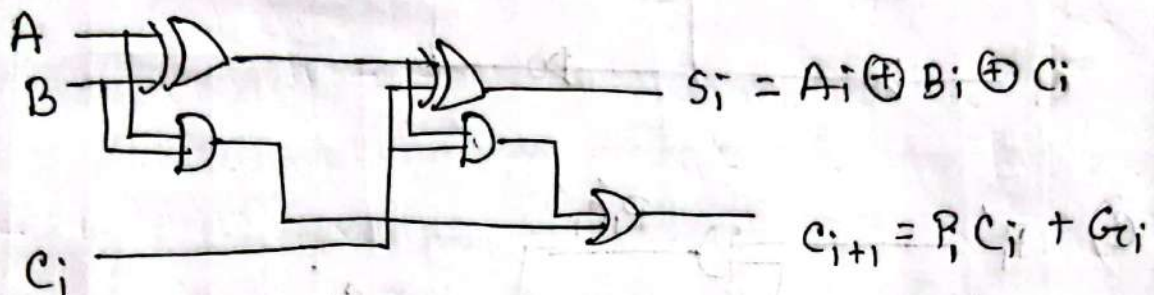
$$= G_2 + P_2 G_1 + P_2 P_1 C_1$$

$$C_4 = G_3 + P_3 C_3$$

$$= G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 C_1$$

$$\begin{aligned}
 C_5 &= G_4 + P_4 C_4 \\
 &= G_4 + P_4 (G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 C_1) \\
 &= G_4 + P_4 G_3 + P_4 P_3 G_2 + P_4 P_3 P_2 P_1 G_1
 \end{aligned}$$

6) Full adder with P & G is shown below



$$C_{i+1} = (C_i G_i' + P_i)'$$

$$\text{R.H.S} = (C_i G_i' + P_i)'$$

$$= (C_i G_i')' \cdot P_i$$

$$= C_i P_i + P_i G_i$$

$$= C_i (A_i + B_i) + (A_i + B_i) A_i B_i$$

$$= C_i P_i + A_i B_i$$

$$= C_i P_i + G_i = C_{i+1} = \text{L.H.S.}$$

Again, we have $S_i = (P_i G_i') \oplus C_i$

$$\text{R.H.S} = P_i G_i' \oplus C_i$$

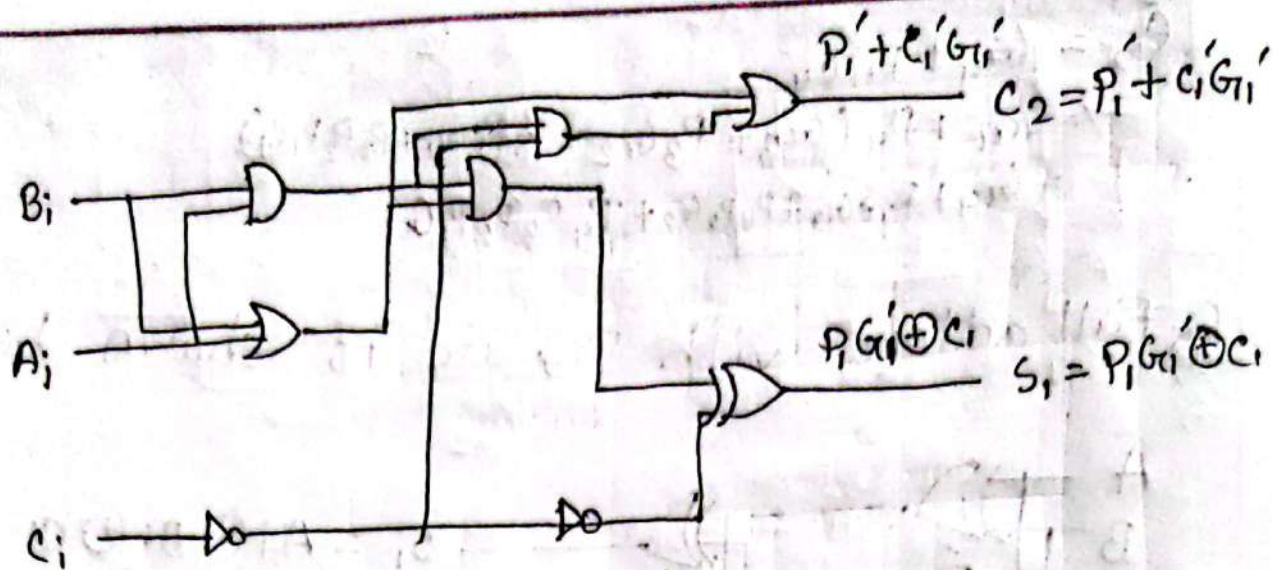
$$= (A_i + B_i) (A_i B_i)' \oplus C_i$$

$$= \{ (A_i + B_i) (A_i + B_i)' \} \oplus C_i$$

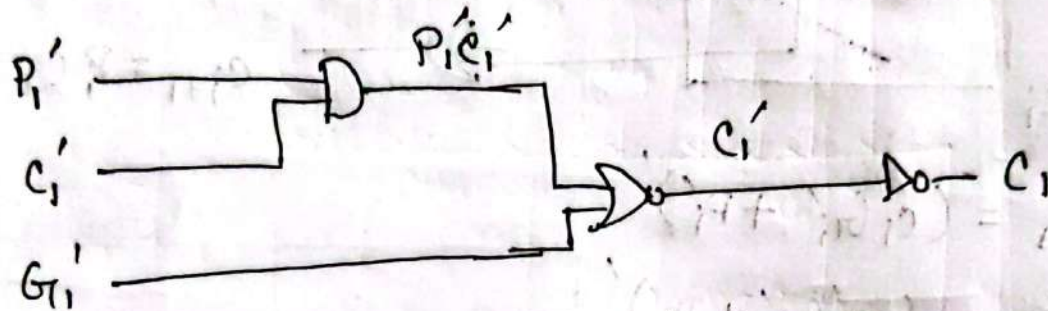
$$= (A_i B_i + A_i B_i') \oplus C_i$$

$$= A_i \oplus B_i \oplus C_i = S_i = \text{L.H.S}$$

b)



c)



$$c_1' = (G_i' + P_i' c_i')'$$

$$= G_i' + (P_i' + c_i')$$

$$= G_i' P_i' + G_i' c_i'$$

$$c_1 = (G_i' P_i' + G_i' c_i')$$

$$c_2 = (G_2' P_2' + G_2' c_2')$$

$$c_3 = (G_3' P_3' + G_3' c_3')$$

$$c_4 = (G_4' P_4' + c_4 G_4')$$

7. a) Here, total delay will be due to three levels. they are -

- i) P and G generator
- ii) Carry generator
- iii) Sum generator

P and G generator has a single level of XOR and AND gates. they will work parallel

$$\text{delay 1} = \max(20, 10) = 20$$

carry generator has two level AND+OR gates

$$\text{delay 2} = 10 + 10 = 20$$

sum generator has single level EX-OR gates

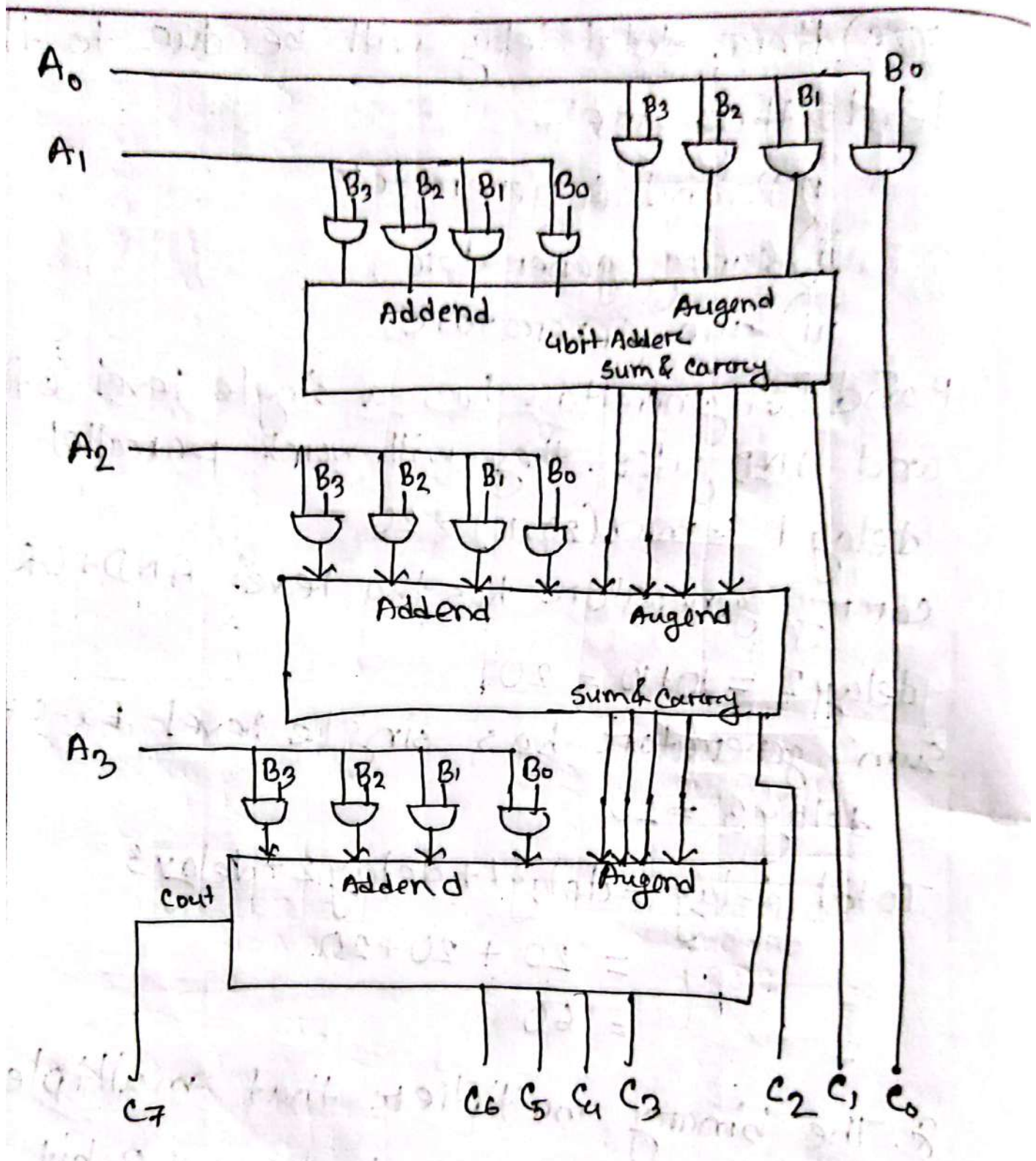
$$\text{delay 3} = 20$$

$$\text{Total delay} = \text{delay 1} + \text{delay 2} + \text{delay 3}$$

$$= 20 + 20 + 20$$

$$= 60$$

8. The binary multiplier that multiplies a 4 bit number $B = b_3b_2b_1b_0$ by a 3 bit number $A = a_2a_1a_0$ to form product $C = c_6c_5c_4c_3c_2c_1c_0$



9. There are in total 512 combinations with a BCD adder. But BCD includes numbers from 0000 through 1001 and hence we have 10 valid combinations for BCD numbers.

Therefore since we have two BCD numbers there are 10×10 valid combinations. Taking carry into consideration there are $10 \times 10 \times 2$ valid combinations. That is there are 200 valid combinations. This implies that there are $512 - 200 = 312$ invalid or don't care conditions. The total number of don't care are 312.

10. the truth table of the combination circuit that generates 9's complement of BCD digit:

BCD				9's complement			
A	B	C	D	W	X	Y	Z
0	0	0	0	-	0	0	0
0	0	0	1	-	0	0	0
0	0	1	0	0	0	1	0
0	0	1	1	0	0	1	0
0	1	0	0	0	0	0	1
0	1	0	1	0	0	0	1
0	1	1	0	0	1	1	1
0	1	1	1	0	1	1	1
1	0	0	0	0	1	0	1
1	0	0	1	0	1	0	1
1	0	1	0	0	1	1	1
1	0	1	1	0	1	1	1
1	1	0	0	0	1	0	1
1	1	0	1	0	1	0	1
1	1	1	0	0	1	1	1
1	1	1	1	0	1	1	1

K-map for w:

AB \ CD	00	01	11	10
00	1	1		
01				
11	x	x	x	x
10			x	x

$W = A'B'C'$

Kmap for x:

AB \ CD	00	01	11	10
00			1	1
01	1	1		
11	x	x	x	x
10			x	x

$X = BC' + B'C$
 $= B \oplus C$

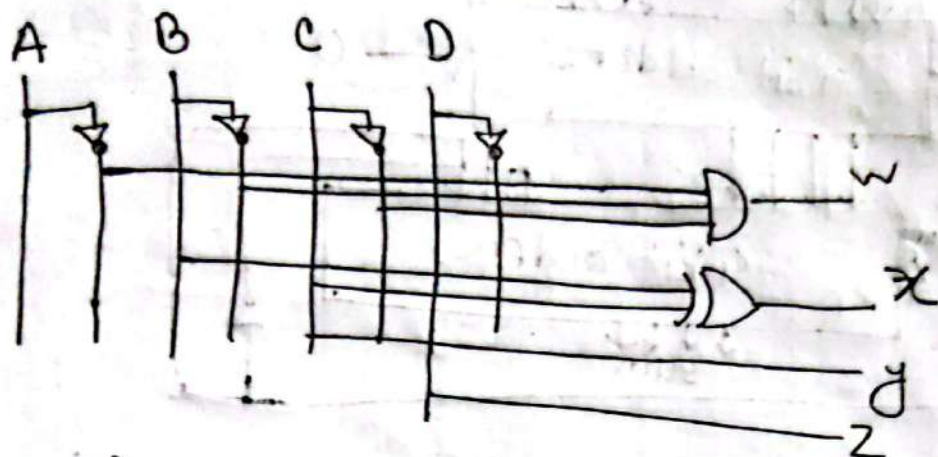
K-map for y:

AB \ CD	00	01	11	10
00			1	1
01			1	1
11	x	x	x	x
10			x	x

$Y = C$

$Z = D'$

Combinational circuit,



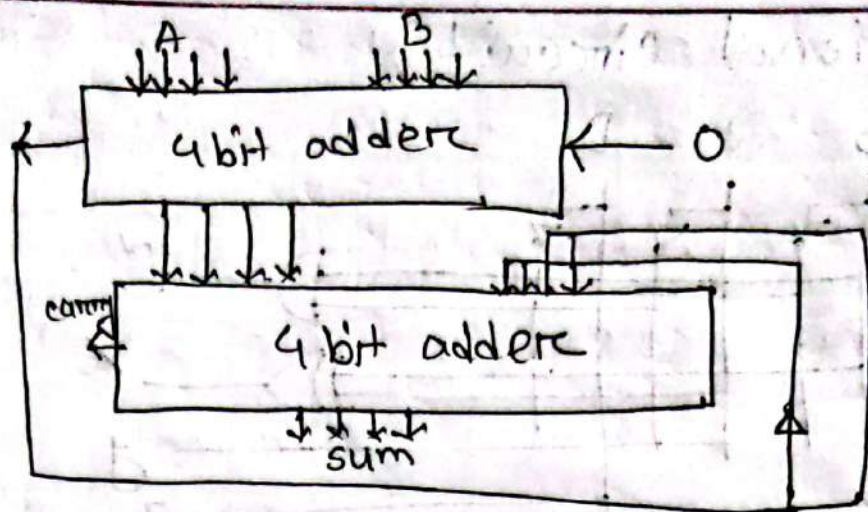
12. a) If the carry is equal to 1, it means that $A+B+6 > 15$, or equivalently $(A+B) > 9$.

The required output carry is therefore the same as the carry from the binary adder.

b) If the output carry is equal to 1 the 4 bit sum in binary is equal to $A+B+6-16$.

$A+B-10$ but required output $A+B+3 = -10$ instead. Hence 0011 (3 in binary) should be added to the 4 bit output.

c) If the output carry is 0 the 4 bit sum in binary is equal to $A+B+6$ but required output $A+B+3$ instead. The correct sum can be produced subtracting 3 from output.



13. Let the two 4 bit numbers are

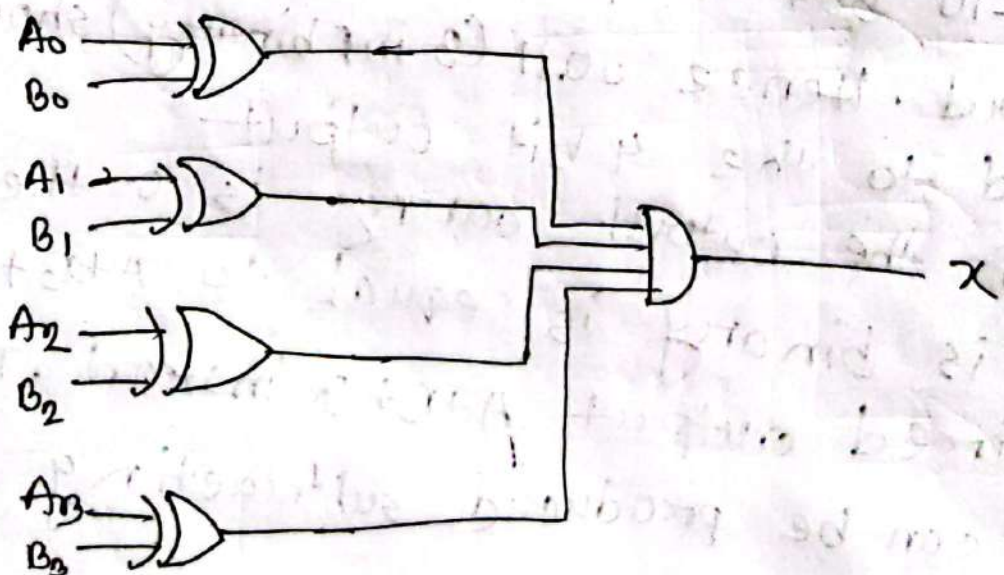
$$A = A_3 A_2 A_1 A_0$$

$$B = B_3 B_2 B_1 B_0$$

Truth Table

A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

X-NOR



Ex 13. Truth table

A	B	C	D	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇	D ₈	D ₉
0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	0	0	1	0	1	0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	0	0	1	0	0	0	0	0	0	0
0	1	0	0	0	0	0	1	0	0	0	0	0	0
0	1	0	1	0	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	0	0	0	1	0	0	0	0
0	1	1	1	0	0	0	0	0	0	1	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0	0	0	0	0
1	0	1	1	0	0	0	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0	0	0	0	0	0	0
1	1	0	1	0	0	0	0	0	0	0	0	0	0
1	1	1	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0	0	0	0

AB \ CD	00	01	11	10
00	D ₀	D ₁	D ₃	D ₂
01	D ₄	D ₅	D ₇	D ₆
11	x	x	x	x
10	D ₈	D ₉	x	x

$$D_0 = A'B'C'D'$$

$$D_1 = A'B'CD'$$

$$D_2 = B'C'D'$$

$$D_3 = B'CD'$$

$$D_4 = BC'D'$$

$$D_5 = B'C'D$$

$$D_6 = BCD'$$

$$D_7 = BCD$$

$$D_8 = AD'$$

$$D_9 = AD$$

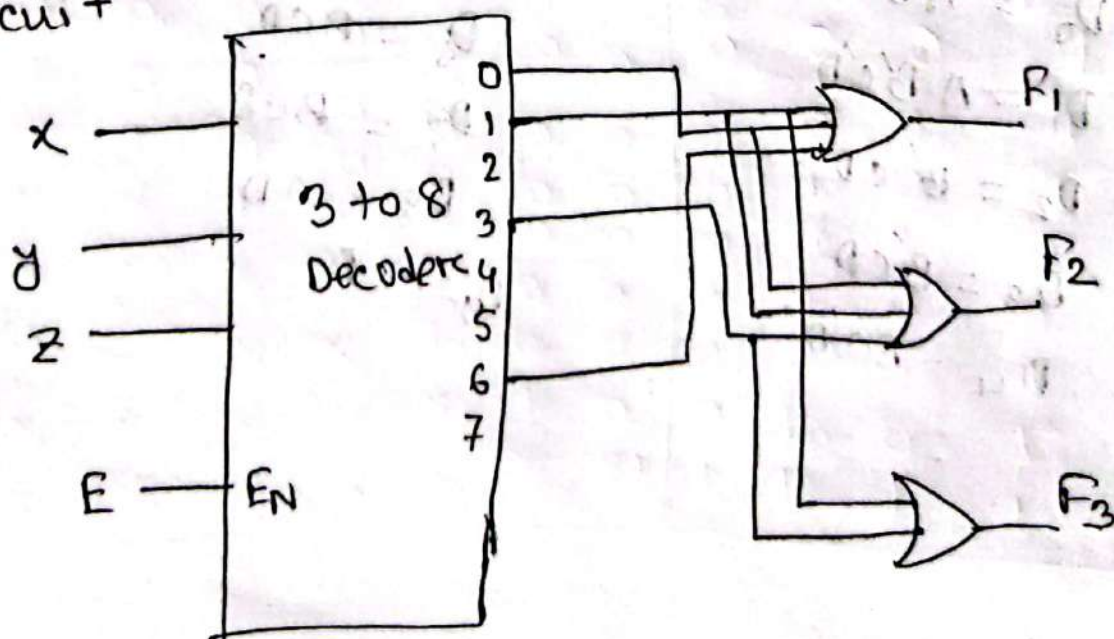
17. The given function are

$$\begin{aligned}
 F_1 &= x'y' + yz \\
 &= x'(z+z')y' + yxz' \\
 &= x'y'z + x'y'z' + xyz' \\
 &= \Sigma(1, 0, 6)
 \end{aligned}$$

$$\begin{aligned}
 F_2 &= x' + y \\
 &= x'(y+y') + y(x+x') \\
 &= x'y + x'y' + xy + x'y \\
 &= x'y + x'y' + xy \\
 &= \Sigma(1, 0, 3)
 \end{aligned}$$

$$\begin{aligned}
 F_3 &= xy + x'y' \\
 &= \Sigma(3, 0)
 \end{aligned}$$

Circuit



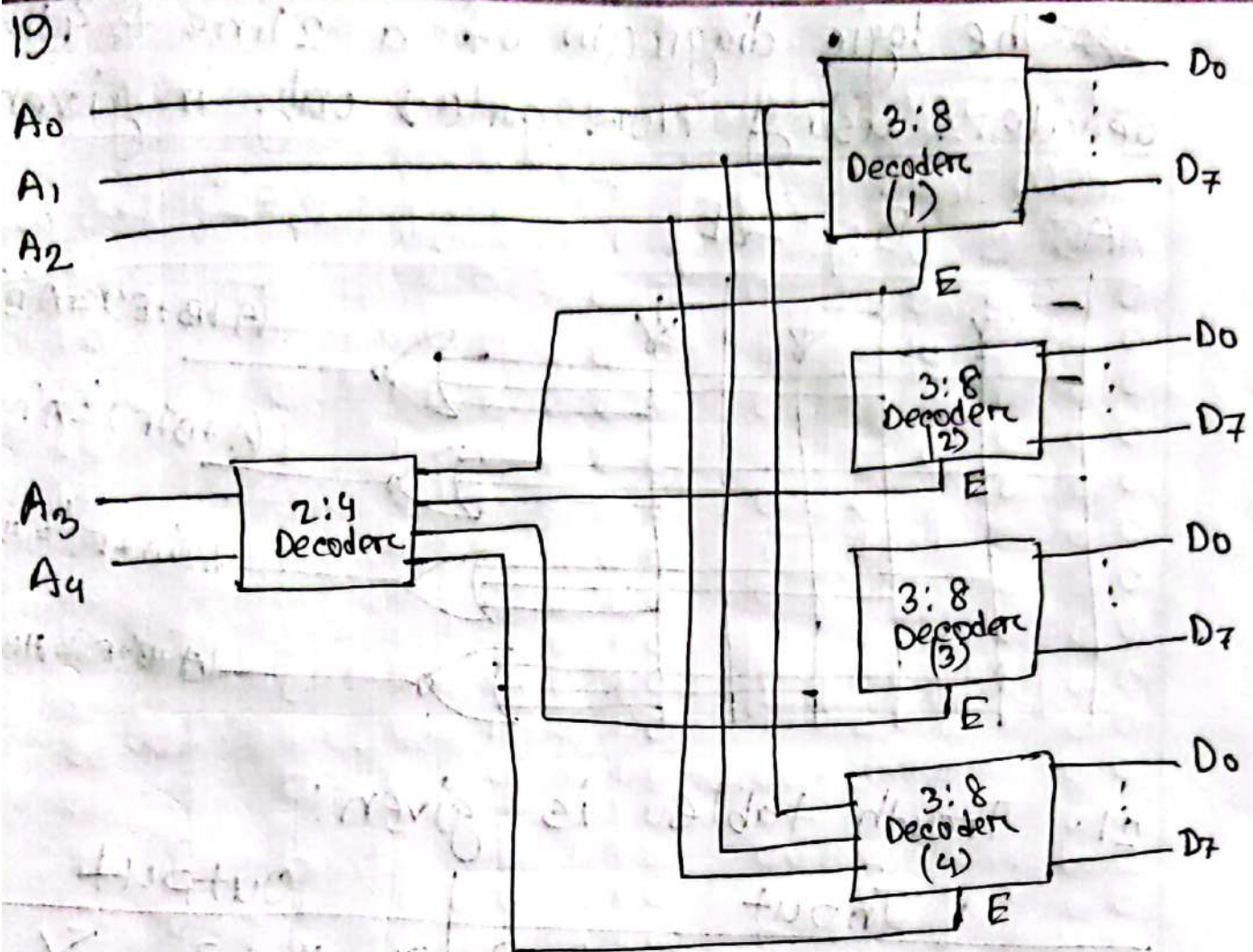
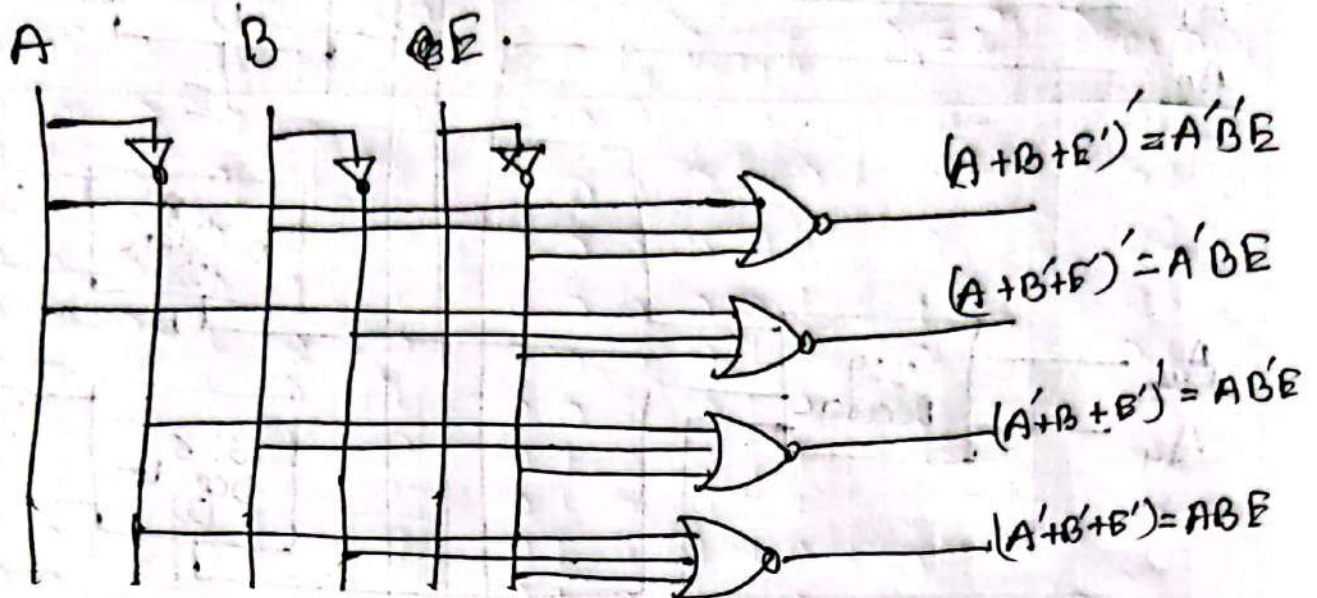


Fig1 5x32 Decoder with four 3x8 decoder and a 2x4 decoder

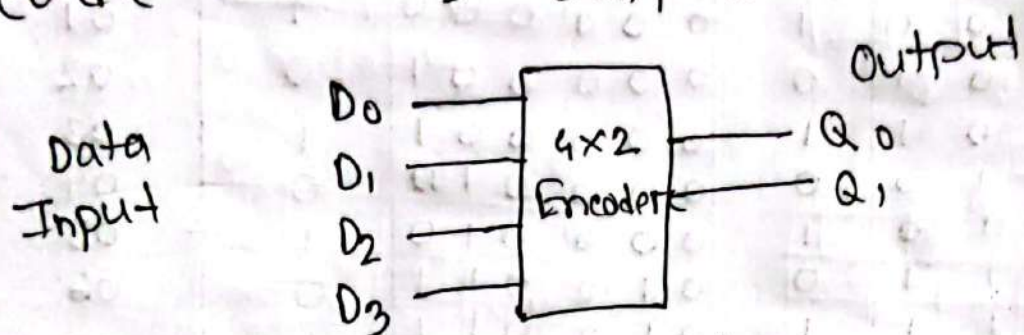
20. The logic diagram of a 2 line to 4 line decoder using NOR gates only is given



21. Truth table is given:

Input								Output			
D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇	x	y	z	v
0	0	0	0	0	0	0	0	x	x	x	0
1	0	0	0	0	0	0	0	0	0	0	1
x	1	0	0	0	0	0	0	0	0	1	1
x	x	1	0	0	0	0	0	0	1	0	1
x	x	x	1	0	0	0	0	0	1	1	1
x	x	x	x	1	0	0	0	0	0	0	1
x	x	x	x	x	1	0	0	0	0	1	1
x	x	x	x	x	x	1	0	0	1	0	1
x	x	x	x	x	x	x	1	1	1	1	1

22. Priority encoders take all of their data inputs one at a time and convert them into an equivalent binary code as its output.



Input				Output	
D ₃	D ₂	D ₁	D ₀	Q ₁	Q ₂
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1
0	0	0	0	X	X

25.

27. Truth table for ROM is given:

ROM Inputs					2^0	Output of ROM						$D_0 = 2^0$	Decimal
A	B	C	D	E		F_6	F_5	F_4	F_3	F_2	F_1		
0	0	0	0	0	0	0	0	0	0	0	0	0	00
0	0	0	0	1	1	0	0	0	0	0	0	1	01
0	0	0	1	0	0	0	0	0	0	0	1	0	02
0	0	0	1	1	1	0	0	0	0	0	1	1	03
0	0	1	0	0	0	0	0	0	0	1	0	0	04
0	0	1	0	1	1	0	0	0	0	1	0	1	05
0	0	1	1	0	0	0	1	0	0	1	1	0	06
0	0	1	1	1	1	0	1	0	0	1	1	1	07
0	1	0	0	0	0	0	1	0	1	0	0	0	08
0	1	0	0	1	1	0	1	0	1	0	0	1	09
0	1	0	1	0	0	0	1	1	0	0	0	0	10
0	1	0	1	1	1	0	1	1	0	0	1	1	11
0	1	1	0	0	0	0	1	1	0	0	1	0	12
0	1	1	0	1	1	0	1	1	0	1	0	1	13
0	1	1	1	0	0	0	1	1	0	1	1	0	14
0	1	1	1	1	1	0	1	1	0	1	1	1	15
1	0	0	0	0	0	0	0	1	1	0	0	0	16
1	0	0	0	1	1	0	0	1	1	0	0	1	17
1	0	0	1	0	0	0	0	1	0	0	0	0	18
1	0	0	1	1	1	0	0	1	0	0	0	1	19
1	0	1	0	0	0	0	0	0	0	0	1	0	20
1	0	1	0	1	1	0	0	0	0	0	1	1	21
1	0	1	1	0	0	0	0	0	0	1	0	0	22
1	0	1	1	1	1	0	0	0	0	1	0	1	23
1	1	0	0	0	0	0	0	0	1	1	0	0	24
1	1	0	0	1	1	0	0	0	1	1	0	1	25
1	1	0	1	0	0	0	0	0	1	0	0	0	26
1	1	0	1	1	1	0	0	0	1	0	0	1	27
1	1	1	0	0	0	0	0	0	0	0	0	0	28
1	1	1	0	1	1	0	0	0	0	0	0	1	29
1	1	1	1	0	0	0	0	0	0	0	1	0	30
1	1	1	1	1	1	0	0	0	0	0	1	1	31
1	1	1	1	1	1	0	0	0	0	0	1	1	32

ROOM Input					2 ⁰	Outputs of ROM						D ₀ =2 ⁰	Decimal
A	B	C	D	E		F ₆	F ₅	F ₄	F ₃	F ₂	F ₁		
1	0	0	0	1	0	0	0	1	0	1	0	0	33
1	0	0	0	1	1	0	0	1	0	1	0	1	34
1	0	0	1	0	0	0	0	1	0	1	1	0	35
1	0	0	1	0	1	0	0	1	0	1	1	1	36
1	0	0	1	0	1	0	0	1	0	1	0	0	37
1	0	0	1	1	0	0	0	1	1	0	0	1	38
1	0	1	0	0	0	1	0	0	0	0	0	0	39
1	0	1	0	0	1	1	0	0	0	0	0	1	40
1	0	1	0	1	0	1	0	0	0	0	1	0	41
1	0	1	0	1	1	1	0	0	0	0	1	1	42
1	0	1	1	0	0	1	0	0	0	1	0	0	43
1	0	1	1	0	1	1	0	0	0	1	0	1	44
1	0	1	1	1	0	1	0	0	0	1	1	0	45
1	0	1	1	1	1	1	0	0	0	1	1	1	46
1	1	0	0	0	0	1	1	0	0	1	0	0	47
1	1	0	0	0	1	1	1	0	0	1	0	1	48
1	1	0	0	1	0	1	0	0	1	0	0	0	49
1	1	0	0	1	1	1	0	0	1	0	0	1	50
1	1	0	1	0	0	1	0	1	0	0	0	0	51
1	1	0	1	0	1	1	0	1	0	0	0	1	52
1	1	0	1	1	0	1	0	1	0	1	0	0	53
1	1	0	1	1	1	1	0	1	0	1	1	1	54
1	1	1	0	0	0	1	1	0	0	1	0	0	55
1	1	1	0	0	1	1	1	0	0	1	0	1	56
1	1	1	0	1	0	1	1	0	0	1	1	0	57
1	1	1	0	1	1	1	1	0	0	1	1	1	58
1	1	1	1	0	0	1	1	1	0	0	0	0	59
1	1	1	1	0	1	1	1	1	0	0	0	1	60
1	1	1	1	1	0	1	1	1	0	0	0	0	61
1	1	1	1	1	1	1	1	1	0	0	1	0	62
1	1	1	1	1	1	1	1	1	1	0	1	1	63

32. Maps for BCD to excess 3 code converter are:

AB \ CD	00	01	11	10
00	1			
01	1			
11	X	X	X	X
10	1		X	X

$$Z = D'$$

AB \ CD	00	01	11	10
00	1			
01	1			
11	X	X	X	X
10	1		X	X

$$Y = CD + C'D'$$

AB \ CD	00	01	11	10
00		1	1	1
01	1			
11	X	X	X	X
10		1	X	X

$$X = B'C + B'D + BC'D'$$

AB \ CD	00	01	11	10
00				
01		1	1	1
11	X	X	X	X
10	1	1	X	X

$$W = A + BC + DB$$

$$W = A + BC + BD$$

$$W' = A'B' + A'C'D'$$

$$X = B'C + B'D + BC'D'$$

$$X' = B'C'D' + BC + BD$$

Programming table:

$$Y = CD + C'D'$$

$$Y' = C'D + CD'$$

$$Z = D'$$

$$Z' = D$$

	Product term	Input				Output			
		A	B	C	D	$F_1(T)$	$F_2(C)$	$F_3(T)$	$F_4(T)$
A	1	1	—	—	—	1	—	—	—
BC	2	—	1	1	—	1	1	—	—
BD	3	—	1	—	1	1	—	1	—
$B'C'D'$	4	—	0	0	0	—	1	—	—
CD	5	—	—	1	1	—	—	1	—
$C'D'$	6	—	—	0	0	—	—	—	1
D'	7	—	—	1	0	—	—	—	1